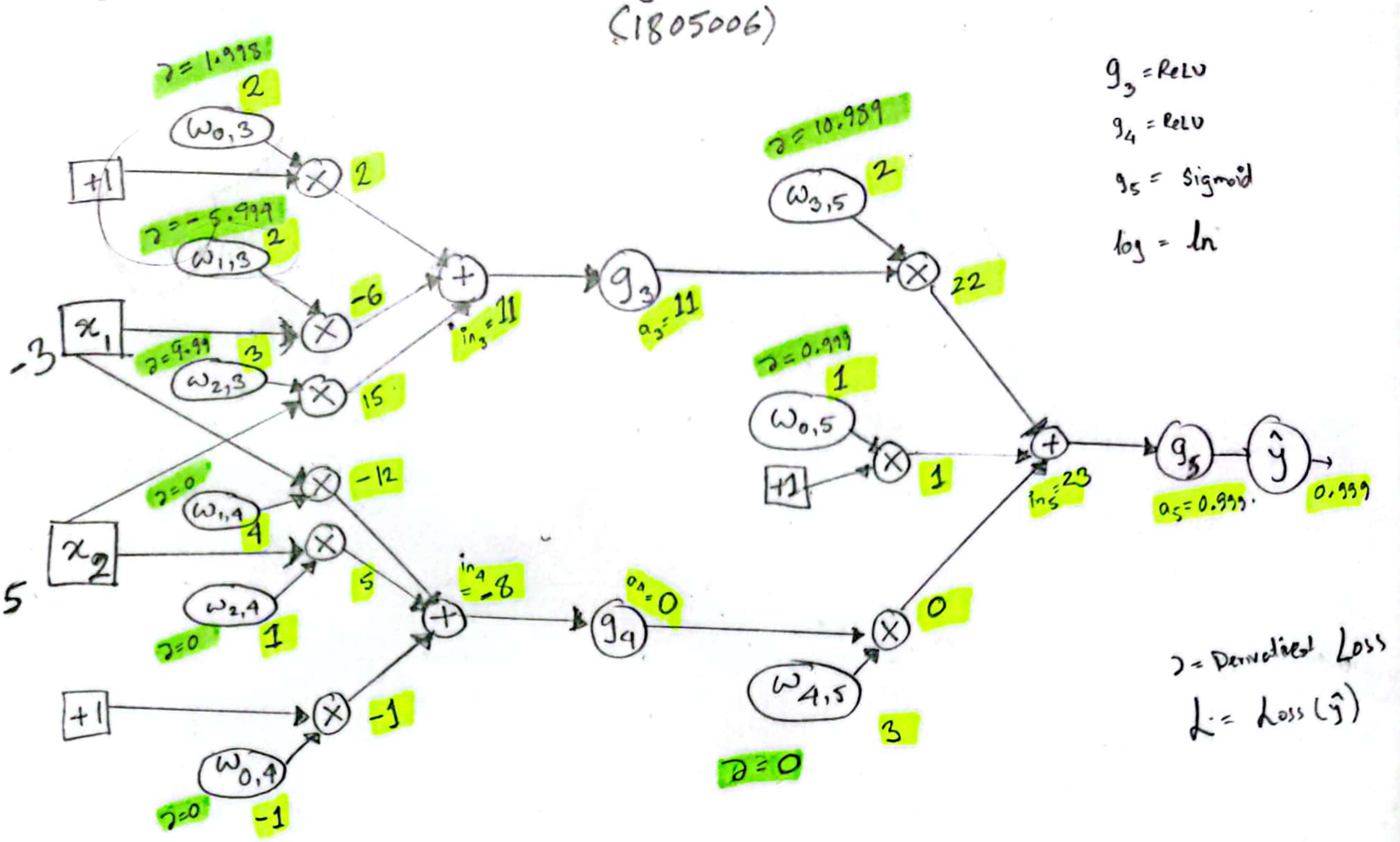
CSE-471 [Machine Learning] Tanjeem Azwad Zaman (1805006) 15-12-23



Forward Propagation:

$in_3 = (w_{0,3} \cdot 1) + (w_{1,3} \cdot x_1) + (w_{2,3} \cdot x_2) = (2)(1) + (2)(-3) + (5)(3) = 11$; $a_3 = g_3(11) = 11$

$in_4 = (w_{0,4} \cdot 1) + (w_{1,4} \cdot x_1) + (w_{2,4} \cdot x_2) = (-1)(1) + (4)(-3) + (5)(1) = -8$; $a_4 = g_3(-8) = -8$

$in_5 = (w_{0,5} \cdot 1) + (w_{3,5} \cdot a_3) + (w_{4,5} \cdot a_4) = (1)(1) + (2)(11) + (3)(0) = 23$; $a_5 = g_5(23) = \frac{1}{1+e^{-23}} = 0.999 = \hat{y}$

Loss (Cross Entropy):

$\mathcal{L}(\hat{y}) = -(y \log \hat{y} + (1-y)\log(1-\hat{y})) = -[0 \cdot \log 0.999 + (1-0)\log(1-0.999)] = 6.91$ || Given, $y = 0$

Backwards Prop:

$g_5'(in_5) = g_5(in_5) \cdot (1 - g_5(in_5)) = (0.999)(1-0.999)$ || $\Delta_5 = \left[\frac{1-y}{1-\hat{y}} - \frac{y}{\hat{y}}\right] \cdot g_5'(in_5) = \left(\frac{1-0}{1-0.999} - \frac{0}{0.999}\right)(1-0.999)(0.999) = 0.999$

$\frac{\partial}{\partial w_{3,5}}(\mathcal{L}(\hat{y})) = \Delta_5 \cdot a_3 = (0.999) \times (11) = 10.989$; $\frac{\partial}{\partial w_{4,5}}(\mathcal{L}(\hat{y})) = \Delta_5 \cdot a_4 = 0.999 \times 0 = 0$; $\frac{\partial}{\partial w_{0,5}}(\mathcal{L}(\hat{y})) = \Delta_5 \times 1 = 0.999$

If $g_i(x)$ is ReLU, $g_i'(x) = \begin{cases} 0 & ; x < 0 \\ 1/2 & ; x = 0 \\ 1 & ; x > 0 \end{cases}$ | $g_3'(in_3) = g_3'(11) = 1$; $g_4'(in_4) = g_4'(-8) = 0$

$\Delta_3 = (\Delta_5)(w_{3,5})(g_3'(in_3)) = (0.999)(2)(1) = 1.998$; $\Delta_4 = (\Delta_5)(w_{4,5})(g_4'(in_4)) = (0.999)(3)(0) = 0$

$\frac{\partial}{\partial w_{0,3}}(\mathcal{L}) = \Delta_3 \cdot 1 = 1.998$; $\frac{\partial}{\partial w_{0,4}}(\mathcal{L}) = \Delta_4 \cdot 1 = 0$

$\frac{\partial}{\partial w_{1,3}}(\mathcal{L}) = \Delta_3 \cdot x_1 = (1.998)(-3) = -5.994$; $\frac{\partial}{\partial w_{1,4}}(\mathcal{L}) = \Delta_4 \cdot x_1 = 0 \cdot (-3) = 0$

$\frac{\partial}{\partial w_{2,3}}(\mathcal{L}) = \Delta_3 \cdot x_2 = (1.998)(5) = 9.99$; $\frac{\partial}{\partial w_{2,4}}(\mathcal{L}) = \Delta_4 \cdot x_2 = (0)(5) = 0$